LAB REPORT

Experiment No: 01
Experiment Name: Introduction to Basic Electronic Components
Course Code: CSE216
Course Title: Electronic Devices and Circuit Lab

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Objectives

1. To become familiar with the basic instruments used in an electronics lab, such as the ammeter, multimeter, oscilloscope, and signal generator, and to learn how to use them.

2. To identify basic electronic components such as the inductor, capacitor, potentiometer, diode, Zener diode, BJT, and MOSFET, and to learn their symbols and construction.

3. To understand the function and characteristics of each component.

4. To measure voltage, current, and resistance using a multimeter, and to identify the terminals and test the condition of diodes and transistors.

5. To observe the waveform from a signal generator on an oscilloscope and determine its frequency and amplitude.

6. To build circuits on a breadboard and develop the habit of using equipment safely.

Introduction

Every electronic circuit we see around us, whether it is a phone charger or a radio, is made from a few basic components. To work with these circuits, we first need to know what each component does and how to check it using the instruments in the lab. That is the main idea behind this experiment.

The components we studied fall into two groups. The first group is passive components. An inductor stores energy in a magnetic field and resists any sudden change in current. A capacitor stores energy in an electric field, and we use it to smooth out voltage or block DC. A potentiometer is a variable resistor, so we can change the resistance by turning a knob. Volume control in a speaker is a common example.

The second group is semiconductor devices. A diode lets current pass in one direction only, so it is used to convert AC to DC. A Zener diode works in reverse breakdown and keeps the voltage almost constant, which is why it is used in voltage regulation. A BJT is controlled by current. A small base current controls a much larger collector current. A MOSFET is controlled by voltage at the gate, and it hardly draws any input current, so it is very good for switching.

We also learned about the instruments used to test these parts. An ammeter measures current. A multimeter can measure voltage, current and resistance, and we can also use it to check diodes and continuity. An oscilloscope shows a signal on the screen as a waveform, so we can find its amplitude and frequency. A signal generator gives us sine, square and triangular waves to use as input for a circuit.

In this experiment we tried to get familiar with all these components and instruments. This knowledge will help us a lot in the later experiments where we build and test real circuits.

Overview of Components and Instruments

1. Ammeter
2. Multimeter
3. Inductor
4. Capacitor
5. Oscilloscope
6. Signal Generator
7. BJT
8. MOSFET
9. Diode
10. Zener Diode
11. Potentiometer

Ammeter

An ammeter is an instrument used to measure the electric current flowing through a part of a circuit. Its unit is the ampere (A), and mA or μA is used for small currents. To measure current, it must always be connected in series with the circuit, so that the whole current passes through the instrument. Its internal resistance is kept very low, otherwise it would change the current in the circuit and give a wrong reading. Because its resistance is almost zero, an ammeter must never be connected directly across a battery or power supply, as the excessive current can damage it. In DC circuits, the polarity must be correct, and the highest range should be selected first and then reduced step by step.



Fig. 1.1: Ammeter

Multimeter

A multimeter is a versatile instrument used to measure voltage, current, and resistance. It combines the functions of an ammeter, a voltmeter, and an ohmmeter in a single device. It can also be used for continuity testing and for checking diodes. To measure voltage, it is connected in parallel with the component, and to measure current, it is connected in series with the circuit. To measure resistance, the circuit power must be turned off and the probes placed across the component. Before use, the selector knob must be set to the correct function (V, A, or Ω) and range, with the red probe in the positive jack and the black probe in the COM jack. Multimeters are of two types, analog and digital, and digital multimeters are more commonly used today.



Fig. 1.2: Multimeter

Inductor

An inductor is a passive component, usually made by winding wire around an iron or ferrite core (air-core inductors also exist). When current flows through it, a magnetic field is created around it and energy is stored in that field. An inductor opposes sudden changes in current, so it passes DC easily but blocks high-frequency AC. Its value is measured in henry (H), and mH or μH is used for smaller values. Inductors are commonly used in filter circuits, power supplies, transformers, and radio tuning circuits. When tested with a multimeter, a good inductor shows very low resistance, and if the wire is broken, there is no continuity.

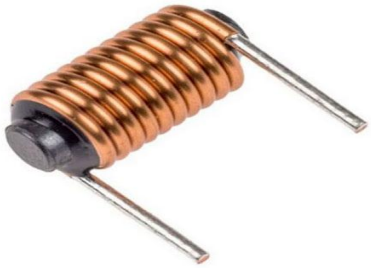


Fig. 1.3: Inductor

Capacitor

A capacitor is a passive component made of two conducting plates separated by an insulating material (dielectric). It can store charge, and therefore energy, in an electric field. A capacitor blocks DC but allows AC to pass, so it is used for DC blocking. Its value is measured in farad (F), though μF, nF, and pF are used in practice. Capacitors are commonly used for smoothing voltage (power supply filters), timing circuits, and signal coupling. Electrolytic capacitors are polarized (+/−), so connecting them in reverse can damage them.

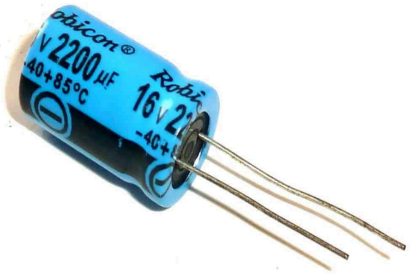


Fig. 1.4: Capacitor

Oscilloscope

An oscilloscope is an instrument that displays the variation of voltage with time as a waveform on its screen. The horizontal axis (X-axis) represents time and the vertical axis (Y-axis) represents voltage. From the waveform, the amplitude, time period, frequency, and shape (sine, square, or triangular) of a signal can be determined. To find the frequency, the time of one complete cycle is first measured using the TIME/DIV setting, and then the formula f = 1/T is used. To find the voltage, the height of the waveform is multiplied by the VOLTS/DIV setting. It usually has channels (CH1, CH2), trigger controls, and probes, and the ground clip of the probe must be connected to the circuit ground. Oscilloscopes are of two types, analog and digital, and digital oscilloscopes can also store waveforms.

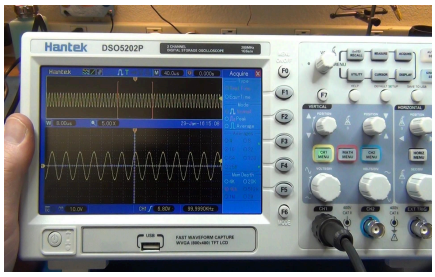


Fig. 1.5: Oscilloscope

Signal Generator

A signal generator is an instrument that produces electrical signals (waveforms) of different shapes and frequencies. It can usually provide sine, square, and triangular waves. Its frequency and amplitude can be changed by turning knobs or pressing buttons. While testing a circuit, this signal is given as the input, and the output is then observed on an oscilloscope. In this way, we can check whether the circuit is working properly. During use, the required waveform, frequency, and amplitude are set first, and then the output terminal is connected to the circuit input. The ground terminal of the instrument must be connected to the circuit ground.



Fig. 1.6: Signal Generator

BJT

A BJT is a three-terminal semiconductor device whose terminals are the base (B), collector (C), and emitter (E). It is made of three semiconductor layers and has two PN junctions. Depending on its construction, it is of two types, NPN and PNP. A BJT is a current-controlled device, meaning a small base current controls a much larger collector current. The current gain is expressed as $\beta = Ic/Ib$. A BJT is used as an amplifier to increase signal strength and as a switch to turn a circuit on and off. Using the diode mode of a multimeter, the base-emitter and base-collector junctions can be tested to identify the terminals and check the condition of the transistor.



Fig. 1.7 : BJT

MOSFET

A MOSFET is a three-terminal semiconductor device whose terminals are the gate (G), drain (D), and source (S). Its gate is separated from the rest of the device by a thin oxide (SiO_2) layer, so almost no current flows into the gate. A MOSFET is a voltage-controlled device, meaning the voltage applied at the gate controls the current flowing from drain to source. It is mainly of two types, N-channel and P-channel, and can also be classified as enhancement mode or depletion mode. It has very high input impedance, fast switching, and low power consumption, so it is widely used in digital circuits, power switching, and amplifiers. A MOSFET can be easily damaged by static electricity, so it must be handled carefully.

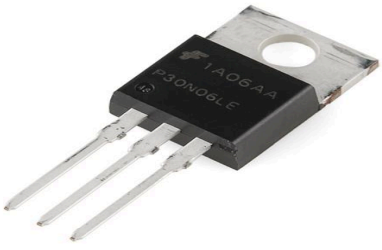


Fig. 1.8 : MOSFET

Diode

A diode is a two-terminal semiconductor device made by joining P-type and N-type semiconductors to form a PN junction. Its two terminals are the anode (P side) and the cathode (N side). In forward bias (positive at the anode, negative at the cathode), the diode allows current to flow, and in reverse bias it almost completely blocks current. A silicon diode starts conducting at about 0.7 V and a germanium diode at about 0.3 V of forward voltage. Because of this one-way conduction property, a diode is used in rectifiers to convert AC to DC. When tested in the diode mode of a multimeter, it shows a small voltage in forward bias and "OL" in reverse bias. A band marked on the diode body indicates the cathode end.



Fig. 1.9 : Diode

Zener Diode

A Zener diode is a specially designed semiconductor diode that conducts current in forward bias like a normal diode, but can also operate safely in reverse bias. When the reverse voltage reaches a certain value, breakdown occurs and current starts to flow. This specific voltage is called the Zener voltage (Vz). After breakdown, the voltage across the diode remains almost constant even if the current increases, and this property is its main function. Therefore, a Zener diode is used as a voltage regulator, and it is connected in reverse bias in parallel with the load. A resistor must be connected in series to limit the current, otherwise excessive current can damage the diode. Zener diodes are available in various voltage ratings, such as 3.3 V, 5.1 V, and 12 V.



Fig. 1.10 : Zener Diode

Potentiometer

A potentiometer is a three-terminal variable resistor whose resistance can be changed by turning a knob or moving a slider. It has a resistive track with two fixed terminals at its ends and a movable slider (wiper) terminal in the middle. The resistance between the two end terminals stays fixed, while moving the wiper changes the resistance between the wiper and either end. It works as a voltage divider, meaning a portion of the input voltage is obtained as the output. Its value is measured in ohms (Ω) or kilo-ohms (kΩ). Potentiometers are used in speaker volume controls, light dimmers, and sensor calibration. When the resistance between the two end terminals is measured with a multimeter, its total value is obtained.



Fig. 1.11 : Potentiometer

Discussion

In this experiment, the basic electronic components and laboratory instruments were studied. The ammeter, multimeter, oscilloscope, and signal generator were introduced, and their functions and methods of connection were discussed. It was learned that the ammeter must be connected in series with the circuit, while the multimeter can be used in different ways to measure voltage, current, and resistance. The oscilloscope was used to observe signal waveforms, and the signals supplied by the signal generator were examined on it. From the displayed waveforms, the amplitude, time period, and frequency were determined.

The passive components, namely the inductor, capacitor, and potentiometer, were also studied. It was observed that the inductor and capacitor store energy in magnetic and electric fields respectively, and that the potentiometer can be used to vary resistance and divide voltage. The semiconductor devices, including the diode, Zener diode, BJT, and MOSFET, were then examined. It was found that the diode allows current in one

direction only, while the Zener diode is used to keep the voltage constant in reverse breakdown. The BJT was identified as a current-controlled device and the MOSFET as a voltage-controlled device. The condition and terminals of the diode and transistor were checked with a multimeter.

During the experiment, it was understood that the instruments must be handled with care. The ammeter was never connected across a voltage source, the polarity of the components was maintained, and the highest range was selected first before the readings were taken. Small errors in the readings may have been caused by the internal resistance of the instruments, loose connections, and the limited accuracy of the scales. Overall, the experiment was found to be helpful in building a foundation for the circuit experiments that will follow.

Conclusion

In this experiment, the basic electronic components and laboratory instruments were studied successfully. The ammeter, multimeter, oscilloscope, and signal generator were introduced, and their functions and proper methods of use were understood. The inductor, capacitor, potentiometer, diode, Zener diode, BJT, and MOSFET were also identified, and their symbols, functions, and characteristics were discussed. It was learned that the instruments must be connected correctly and handled with care to obtain accurate readings and to avoid damage. The knowledge that was gained from this experiment will be helpful in building and analyzing circuits in the experiments that follow.